Answer Key & Marking Scheme

Subject: Mathematics | **Test:** Test 1 (Day 7, 16 May 2026) — Weekly
Maximum Marks: 80

Syllabus: Real Numbers • Polynomials • Pair of Linear Equations in Two Variables • Quadratic Equations • Arithmetic Progressions • Triangles

SECTION A — Answer Key (1 mark each)

Q1. (b) 4 — Prime factorisations: 96 = 2■·3, 404 = 2^2·101, common = 2^2 = 4.

Q2. (b) xy^2 — Take the lowest power of each common prime.

Q3. (c) 3 — Denominator after removing common factors is of form $2^3 \cdot 5^2$; the larger of (3,2) gives 3 decimal places.

Q4. (c) 2 — A quadratic polynomial has at most 2 zeroes (degree 2).

Q5. (a) 5 — $\alpha + \beta = -(-5)/1 = 5$.

Q6. (c) infinitely many solutions — a■/a■ = b■/b■ = c■/c■ = 1/2.

Q7. (b) intersecting — A unique solution corresponds to one point of intersection.

Q8. (a) 5, –2 — $(x - 5)(x + 2) = 0$.

Q9. (b) –8 — $D = b^2 - 4ac = 16 - 24 = -8$.

Q10. (b) zero — $D = 0 \Rightarrow$ equal real roots.

Q11. (a) 47 — $a + 9d = 2 + 9 \cdot 5 = 47$.

Q12. (c) –1 — Difference between consecutive terms is –1.

Q13. (b) 210 — $S = 20 \cdot 21/2 = 210$.

Q14. (a) $\Delta PQR \sim \Delta CAB$ — Match the sides AB/QR ↔ BC/PR ↔ CA/PQ ⇒ corresponding vertices P–C, Q–A, R–B.

Q15. (b) 2 cm — By BPT, AD/DB = AE/EC ⇒ 1.5/3 = 1/EC ⇒ EC = 2.

Q16. (b) proportional — SSS similarity criterion.

Q17. q = 13 — $65 = 5 \cdot 13 + 0$.

Q18. Irrational.

Q19. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

Q20. $\alpha\beta = 6$ — Product of zeroes = c/a = 6/1 = 6.

SECTION B — Answer Key (2 marks each)

Q21. 26 = 2·13, 91 = 7·13. HCF = 13. LCM = 2·7·13 = 182. Verification: HCF × LCM = 13 × 182 = 2366 = 26 × 91. ✓

Q22. Required polynomial = $x^2 - (\text{sum})x + (\text{product}) = x^2 - (3 + (-2))x + (3\cdot(-2)) = x^2 - x - 6$. Verification: zeroes are obtained by factorising $(x - 3)(x + 2) = 0 \Rightarrow x = 3, -2$. ✓

Q23. Subtract: $(2x + 3y) - (2x - 4y) = 11 - (-24) \Rightarrow 7y = 35 \Rightarrow y = 5$. Substitute: $2x + 15 = 11 \Rightarrow x = -2$. So **x = –2, y = 5**.

Q24. $2x^2 - 7x + 3 = 2x^2 - 6x - x + 3 = 2x(x - 3) - 1(x - 3) = (2x - 1)(x - 3)$. Roots: x = 1/2, 3.

Q25. By BPT, AD/DB = AE/EC ⇒ 4/6 = 5/EC ⇒ EC = 30/4 = 7.5 cm.

SECTION C — Answer Key (3 marks each)

Q26. *Proof (by contradiction).* Assume $\sqrt{5} = p/q$ (p, q coprime integers, $q \neq 0$). Then $5q^2 = p^2 \Rightarrow 5 \mid p^2 \Rightarrow 5 \mid p$. Let p = 5k. Then $5q^2 = 25k^2 \Rightarrow q^2 = 5k^2 \Rightarrow 5 \mid q$. So 5 divides both p and q, contradicting that they are coprime. Hence $\sqrt{5}$ is irrational. (3)

Q27. Polynomial division: $x^3 - 3x^2 + 5x - 3 \div (x^2 - 2)$. First term: x. Multiply: $x(x^2 - 2) = x^3 - 2x$. Subtract: $(-3x^2 + 5x - 3) - (-2x) = -3x^2 + 7x - 3$. Next term: -3. Multiply: $-3(x^2 - 2) = -3x^2 + 6$. Subtract: $(7x - 3) - (6) = 7x - 9$. **Quotient = $x - 3$, Remainder = $7x - 9$.**

Q28. Let speed of boat = u km/h, stream = v km/h. Let $1/(u - v) = a$, $1/(u + v) = b$. Then $30a + 44b = 10$ and $40a + 55b = 13$. Solving: multiply first by 4 and second by 3 to get $120a + 176b = 40$ and $120a + 165b = 39 \Rightarrow 11b = 1 \Rightarrow b = 1/11 \Rightarrow u + v = 11$. Then $30a + 44/11 = 10 \Rightarrow 30a = 6 \Rightarrow a = 1/5 \Rightarrow u - v = 5$. Solving: **$u = 8$ km/h, $v = 3$ km/h.**

Q29. Let the integers be x and x + 2. Then $x(x + 2) = 323 \Rightarrow x^2 + 2x - 323 = 0 \Rightarrow (x + 19)(x - 17) = 0$. Since x is positive odd, $x = 17$. The integers are **17 and 19**.

Q30. $a = 8$, $d = -5$, $n = 22$. S■ = $(n/2)(2a + (n-1)d) = (22/2)(16 + 21\cdot(-5)) = 11 \times (16 - 105) = 11 \times (-89) =$ **-979**.

Q31. *Proof (Apollonius).* Drop perpendicular AE from A to BC (extended if necessary). Let BD = DC = m. Using Pythagoras in ΔABE: $AB^2 = AE^2 + BE^2$. In ΔACE: $AC^2 = AE^2 + CE^2$. And in ΔADE: $AD^2 = AE^2 + DE^2$. Since $BE = BD - DE = m - DE$ and $CE = m + DE$: $AB^2 + AC^2 = 2AE^2 + (m - DE)^2 + (m + DE)^2 = 2AE^2 + 2m^2 + 2DE^2 = 2(AE^2 + DE^2) + 2m^2 = 2AD^2 + 2BD^2$. ✓

SECTION D — Answer Key (5 marks each)

Q32. Apply Euclid's algorithm: $12576 = 4052\cdot3 + 420$; $4052 = 420\cdot9 + 272$; $420 = 272\cdot1 + 148$; $272 = 148\cdot1 + 124$; $148 = 124\cdot1 + 24$; $124 = 24\cdot5 + 4$; $24 = 4\cdot6 + 0$. So HCF = **4**. Back-substituting: $4 = 124 - 24\cdot5 = 124 - 5\cdot(148 - 124) = 6\cdot124 - 5\cdot148 = 6\cdot(272 - 148) - 5\cdot148 = 6\cdot272 - 11\cdot148 = 6\cdot272 - 11\cdot(420 - 272) = 17\cdot272 - 11\cdot420 = 17\cdot(4052 - 9\cdot420) - 11\cdot420 = 17\cdot4052 - 164\cdot420 = 17\cdot4052 - 164\cdot(12576 - 3\cdot4052) = 509\cdot4052 - 164\cdot12576$. So **$m = 509$, $n = -164$**. (5)

Q33. Plot tables for $x + 2y = 6$ (eg (0,3), (6,0), (2,2)) and $2x - y = 2$ (eg (0,–2), (1,0), (2,2)). Lines intersect at (2, 2). x-intercepts: (6,0) and (1,0). The triangle formed by the two lines and the x-axis has vertices A(1,0), B(6,0) and C(2,2). Base AB = 5 units, height (vertical distance from C to x-axis) = 2 units. Area = $\frac{1}{2} \times 5 \times 2 =$ **5 sq. units**. (5)

Q34. Let Rehman's present age = x years. Then $1/(x - 3) + 1/(x + 5) = 1/3$. $\Rightarrow [(x + 5) + (x - 3)] / [(x - 3)(x + 5)] = 1/3 \Rightarrow 3(2x + 2) = x^2 + 2x - 15 \Rightarrow 6x + 6 = x^2 + 2x - 15 \Rightarrow x^2 - 4x - 21 = 0 \Rightarrow (x - 7)(x + 3) = 0 \Rightarrow x = 7$ (rejecting –3). **Rehman's present age = 7 years.** (5)

Q35. *Statement of BPT:* If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.
Proof: In ΔABC, draw DE ■ BC. Join BE and CD. Drop perpendiculars $EM \perp AB$ and $DN \perp AC$. Now: $ar(\Delta ADE)/ar(\Delta BDE) = (\frac{1}{2}\cdot AD\cdot EM)/(\frac{1}{2}\cdot DB\cdot EM) = AD/DB$. Similarly $ar(\Delta ADE)/ar(\Delta CDE) = AE/EC$. But ΔBDE and ΔCDE are on the same base DE and between the same parallels DE ■ BC, so $ar(\Delta BDE) = ar(\Delta CDE)$. Hence $AD/DB = AE/EC$. ✓
Application: Given $AD/DB = AE/EC = 2/3$, so $AD : AB = 2 : 5$ and similarly $AE : AC = 2 : 5$. By the converse of BPT (or by similarity $\Delta ADE \sim \Delta ABC$), $DE/BC = AD/AB = 2/5$. So $BC = (5/2)\cdot DE = (5/2)\cdot4 =$ **10 cm**. (5)

SECTION E — Answer Key (4 marks each)

Q36. (i) HCF(24, 36): $24 = 2^3\cdot3$, $36 = 2^2\cdot3^2$. HCF = $2^2\cdot3$ = **12**. (1)
(ii) 240 / 24 = **10 boxes**. (1)
(iii) $12 = 36 - 24$, i.e. $12 = 24\cdot(-1) + 36\cdot(1)$. (2)
OR LCM(24, 36) = $2^3\cdot3^2 = 72$. Verify: HCF·LCM = $12\cdot72 = 864 = 24\cdot36$. ✓ (2)

Q37. (i) Length = (x + 5) m; equation: $x(x + 5) = 750 \Rightarrow$ **$x^2 + 5x - 750 = 0$**. (1)
(ii) $x^2 + 5x - 750 = 0 \Rightarrow (x - 25)(x + 30) = 0 \Rightarrow x = 25$ (rejecting –30). **Breadth = 25 m**. (2)
(iii) Length = 30 m. Perimeter = 2(25 + 30) = **110 m**. (1)

Q38. (i) **Yes — by AA similarity.** Sun's rays make the same angle with the ground, and both objects are vertical, so corresponding angles are equal. (1)
(ii) Friend's height / Friend's shadow = Tree's height / Tree's shadow $\Rightarrow 1.5 / 1 = h / 8$. (1)
(iii) $h = 1.5 \times 8 =$ **12 m**. (2)

— End of Answer Key —